Tactix H2 proof transaction checkpoint

Codex for Andrew

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Outcome. The original 42-clause H2 public-action replay now produces a native DSR proof that verifies through the canonical no-option `dsr-trim` route in both forward and backward directions. Pinned Trestle independently verifies both resulting LSRs against the original CNF. A Release replay and an ASan+UBSan replay emitted identical proof bytes. This is a focused proof-emission repair; it does not close N2 or any N3/N4 performance gate.

Cause and change. A Backbone failed-literal action emitted the first H2 root unit, and a later legal CCE action deleted clauses near it. The earlier native DSR writer had no record of the unit's RUP support. Binary clause IDs cannot supply that record: native DSR intentionally drops them. The candidate captures the exact clause literals in the failed-literal implication cone before unwinding speculation. The proof writer delays the named unit, writes later deletions of clauses outside that cone first, then writes the unit and the support deletions. It spools delayed deletion bytes to an anonymous file, avoiding a deletion-batch memory ledger. The public X7 action and its legal instances remain unchanged; no checker option or contract changed.

Checks. The exact public replay passed both checker directions and both original-CNF Trestle checks in Release job 1723729 and ASan job 1723764. Both emitted DSR SHA-256 `4075e6c59eede852dd487f0ee7ce23a97e1622e970e7759b99a7f67568288579`. The output puts 24 non-support deletions before the first unit and retains its five immediately deleted antecedents afterward. The frozen-checker focused Release suite passed 7/7; focused ASan+UBSan passed 4/4. Additional Release and ASan serializer tests passed for the next-addition and self-deletion transaction boundaries. Current-source full Debug job 1723753 is running; Opus-owned affected fuzz is live on a read-only source patch and binary pin.

An initial `micro_proofs` run used the repository's known stale local checker clone and rejected corpus-034 under its extra every-lemma audit. The candidate and prior solver emitted byte-identical proofs for that case; the frozen checker pin passed the complete focused suite. This was not counted as a solver regression or used to relax checking.

Readiness and next actions. The AM446 source audit still finds a general live pseudo-Boolean proof-interface gap relative to LymphoSAT/Satsuma winner paths. The current online strategy surface can prototype inprocessing choices, but production learning still needs an adjustable inprocessing controller and owned live switches for branching and clause deletion. Complete the current Debug and Opus fuzz gates, repair any bounded failure, and run the combined independent proof/model and N1–N4 performance/reference gates before deciding readiness. The 1,166-case Tier D comparison remains after verified goal completion, at 5,000 seconds, four CPUs and 32 GiB per instance.

Evidence. benchmarks/reports/2026-09-25-am452-h2-dsr-unit-transaction.md; benchmarks/reports/2026-09-25-am450-h2-emitter-boundary.md; benchmarks/reports/2026-09-25-am449-h2-rup-support-order-control.md; benchmarks/reports/2026-09-25-am446-reference-proof-producer-crosswalk.md.